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When macro time series meets micro panel data: A clear and present danger[☆]

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ABSTRACT

This short note addresses a serious problem in combining macro (country-level) time series data, e.g., oil price uncertainty (OPU) or economic/trade policy uncertainty (EPU/TPU), with micro (firm-level) financial and accounting panel data, e.g., corporate leverage, investment, and innovation, to name a few. In most of the applications, the main interest is to assess the impacts of country-level explanatory variable on the firm-level dependent variable, with year fixed effects (along with other firm fixed effects, etc.) being included. Since the macro time series are the same for all firms in each year, it is straightforward to show that the macro time series variable is *perfectly correlated* with the year fixed effects, and thus unidentifiable. We employ three real data sets to illustrate the perfect multicollinearity issue, and our demonstrations cast doubt on the findings of several existing studies suffering from this issue. Finally, we also offer some practical ways to get around with (at least mitigate) this problem.

1. Introduction

Over the past few years, there are increasing empirical studies investigating the effects of key macro-level time series variables (oil price uncertainty, economic policy uncertainty, or trade policy uncertainty) on the micro-level firms' panel data variables (corporate leverage, investment, or innovation), e.g., Fan et al. (2021), Hou et al. (2021) and Shen and Hou (2021), to name a few.[1] In part or all of the analysis, authors include year-specific dummy variables (along with firm-specific dummy variables) to control for nation-wide, macroeconomic shocks and trend that shape micro-level firm dependent variables. The main purpose of this article is (i) to point out that there is a perfect multicollinearity issue in such settings that renders the coefficient of the key macro-level (time series) variable unidentifiable, and (ii) to offer some possible practical ways to get around with this problem.

First, we illustrate the (perfect multicollinearity) issue using three empirical articles published in this Journal. The idea is that, in a standard (micro) firm-level panel data model, researchers often employ the fixed effects (FE) estimators to obtain consistent estimates of variables of key interests. In general, individual (firm) FEs and time (year) FEs are included to account for the possible impacts of unobserved firm-level and macroeconomic-level factors. However, if one's interest is in the effect of macro-level variable, one needs to combine micro-level panel data with macro-level (time series) variable. As explained later, it is straightforward to see that the macro-level time series variable is perfectly correlated with year dummy variables. As a result, the parameter of the macro-level variable is not identifiable.

Second, we proceed to offer some practical directions to get around (or, alleviate) this commonly made mistake. Admittedly, every approach has its own benefits and disadvantages.[2] Those methods include (i) ignoring year dummies, (ii) constructing individual-variant macro-level variables, i.e., panel rather than time series data, (iii) building differential micro-level intensity/sensitivity data and utilizing the (generalized) difference-in-differences approach of Rajan and Zingales (1998), and/or (iv) performing cross-country analysis.

This article is organized as follows. Section 2 demonstrates the existence of perfect multicollinearity among macro-level time series data and the year dummies. Section 3 replicates three recent papers published in this Journal. Section 4 provides some preliminary suggestions for the future research. Section 5 concludes.

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[1] We thank a referee for suggesting that some other possible/potential applications of important macro time series variables (on micro panel data) include monetary policy uncertainty (e.g., Creal and Wu, 2017; Kurov and Stan, 2018), geopolitical risk uncertainty (e.g., Oanh and Hoang, 2021; Shen et al., 2021; Wang et al., 2021), and exchange rate uncertainty (Binding and Dibiasi, 2017; Li et al., 2019).

[2] Obviously, it seems no *perfect* solution to this problem.

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2. Perfect multicollinearity

Consider a simple but representative data set, which is composed by a micro panel data set and a macro time series data set. For the micro part, suppose that there are four firms in a country, i.e., id = 1, 2, 3, and 4. For each firm, there are four annual observations for year = 2011, 2012, 2013, and 2014, respectively. The firm-level dependent variable is y and the (only) firm-level explanatory variable is x. For the macro part, assume that the country's gross domestic products (GDP) for these four years are 5.3, 6.2, 7.5 and 7.8, respectively. The main interest (of many papers) is to evaluate the effect of (macro-level independent variable) GDP on (firm-level dependent variable) y, along with other controlling variables.[3] Note that, since we are considering a panel data model, it is common practice to include firm fixed effects (or dummies) to control for unobserved, time-invariant firm-specific characteristics that affect y across firms. More related to the focus of this paper, it is also standard to include year fixed effects (or dummies) to control for nation-wide shocks and trends that influence y over time.

The problem with this regression is that the main explanatory variable GDP_t is perfectly correlated with four year dummies year2011_t, year2012_t, year2013_t and year2014_t, where year2011_t (the other three year dummies are defined similarly) is equal to 1 if the observation is in year 2011, and 0 otherwise. To see this,

$$\text{GDP}_t = 5.3\times\text{year2011}_t + 6.2\times\text{year2012}_t + 7.5\times\text{year2013}_t + 7.8\times\text{year2014}_t$$

Clearly, GDP_t is an exact, linear combination of year dummies, i.e., there is perfect multicollinearity. As a consequence, the regression coefficients are indeterminate, including the interested coefficient on GDP_t (apart from year dummy variables). Unfortunately, this is precisely the situation that many published articles encounter, and the unidentification of key parameters renders their findings and conclusions questionable, if not meaningless.

So, the next question is: how do the authors estimate and report the (senseless) results under this situation? So far as we can see, most articles use Stata software to perform estimation, and Stata will automatically omit variables that cause the problem of perfect multicollinearity (until there are no perfectly collinear covariates).[4] However, there are few people aware of how Stata deals with the perfect multicollinearity issue, and there seems no deterministic rule for Stata to delete (perfectly) collinear variable(s) in case of perfect multicollinearity. Accordingly, *the estimation results are likely to be different depending on which variables are dropped.* In the next section, we employ three articles recently published in this Journal to demonstrate this issue.

3. Replications of three papers

In the following, we utilize the data/code provided by Fan et al. (2021), Hou et al. (2021), and Shen and Hou (2021) to demonstrate the consequences of perfect multicollinearity. Note that, for brevity, we only briefly explain the main variables (the dependent variables and the macro-level time series variable) and refer the readers to the original articles for more details on the controlling variables.

3.1. Empirical specifications

In the first paper, Fan et al. (2021) rely upon the following specification to assess the relationship between corporate leverage and oil price uncertainty, and report the results in their Table 3 (p. 5):

$$\text{Leverage}_{i,t} = \alpha + \beta\,\text{OVX}_{t-1} + \gamma\,\text{Controls}_{i,t-1} + \eta_i + \kappa_t + \epsilon_{i,t} \tag{1}$$

where $\text{Leverage}_{i,t}$ (LEV) represents book leverage of firm i in year t, and the key (macro-level) explanatory variable OVX_{t-1} (OVX) denotes one-year lagged oil price uncertainty (OPU). Moreover, Fan et al. (2021) also consider two alternative dependent variables, namely, Short_LEV (short-term leverage) and Long_LEV (long-term leverage). Note that the parameter on OVX, i.e., β, is their major interest and measures the extent to which oil price uncertainty affects corporate leverage. Firm fixed effect η_i and year fixed effect κ_t are included to alleviate the possible influences of unobserved firm-level and macroeconomic factors, respectively.

In the second paper, Hou et al. (2021) evaluate the effect of economic policy uncertainty (macro time series) on investment inefficiency (micro panel data) of energy and power firms. Their empirical model is:

$$INEINV_{i,t} = \alpha_0 + \alpha_1\,EPU_{i,t-1} + \gamma\text{Controls}_{i,t-1} + YearFE + \mu_{i,t} \tag{2}$$

where $INEINV_{i,t}$ is the dependent variable, which denotes corporate investment inefficiency obtained from the method of Richardson (2006). The primary explanatory variable is economic policy uncertainty $EPU_{i,t-1}$ of Baker et al. (2016). Apart from other controlling covariates, $YearFE$, i.e., year fixed effects, are also included.

In the third paper, Shen and Hou (2021) explore the impact of trade policy uncertainty (macro time series) on corporate innovation (micro panel data) in new energy vehicle industry. Their empirical regression is:

$$RDS_{i,t}/PAT_{i,t} = \alpha_0 + \alpha_1\,TPU1_{i,t-1} + \gamma\text{Controls}_{i,t-1} + Year_t + \mu_{i,t} \tag{3}$$

where the dependent variable is either $RDS_{i,t}$ or $PAT_{i,t}$, which represents corporate innovation. Shen and Hou (2021) define the input of innovation (RDS) as ratio of research and development expenditures divided by total assets multiplied by 100. In addition, they measure the output of innovation (PAT) by the number of granted patents. Two patent-based metrics are considered: (i) the natural logarithm of one plus a firm's total number of patents, including invention patents, utility model patents and design patents (PAT1), and (ii) the natural logarithm of one plus a firm's total number of invention patents and utility model patents (PAT2). The key explanatory variable of interest is trade policy uncertainty $TPU1_{i,t-1}$.[5] Finally, in addition to the control variables, the regression includes the year fixed effects, i.e., $Year_t$.

3.2. Empirical results

In the (unpublished) online appendix, we replicate the main results of these three articles in Table A1, A2, and A3, respectively. Moreover, we selectively drop different pairs of year dummies and report contrasting outcomes in the same Table. The general message from these Tables is that one might obtain exactly opposite (in terms of signs) conclusions by removing alternative year dummies so as to deal with the perfect multicollinearity problem.

To gain a more thorough understanding of the distribution of possible macro-level (OVX, EPU, and TPU1) coefficients, we re-run Eq. (1), (2), and (3) sequentially by alternatively omitting all possible pairs of year dummies. After dropping duplicated estimates, there are (1) 55 unique estimated values on OVX, (2) 66 distinct estimates on EPU, and (3) 66 unique observations on TPU1, respectively. Figs. 1 to 3 display the corresponding bar graphs for the distribution of all estimates.

Fig. 1 displays the bar graphs for the distribution of all OVX estimates when alternative dependent variables (LEV, Short_LEV, and

[3] This toy example is available upon request.

[4] While it is generally observed that Stata omits variables starting from the end of the variable list (in the Stata command line) and working its way forward until perfect collinearity has been resolved, this is not always true under all situations.

[5] As in Hou et al. (2021), a better way is to denote trade policy uncertainty by $TPU1_{t-1}$, i.e., without subscript i, since the TPU index is NOT individual-variant for each year t.

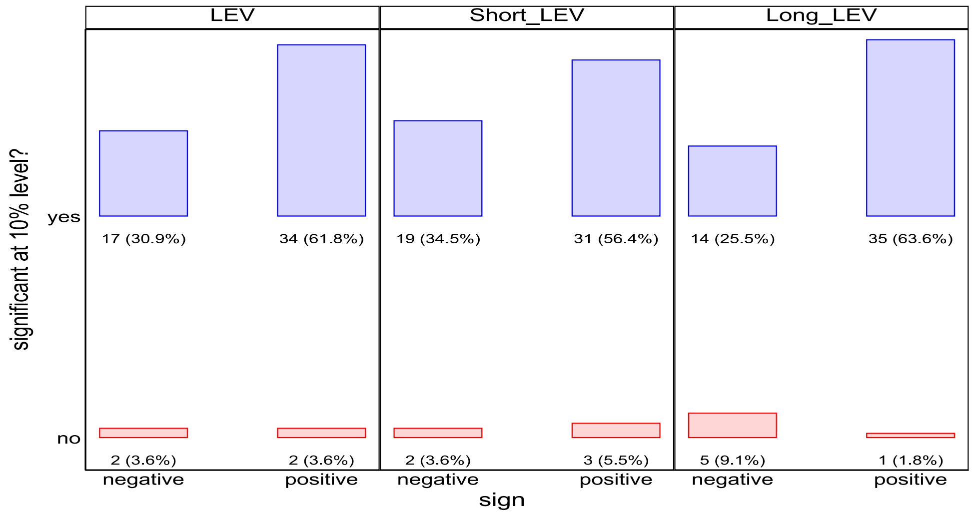

Fig. 1. Estimates of OVX on LEV, Short_LEV, and Long_LEV, respectively.

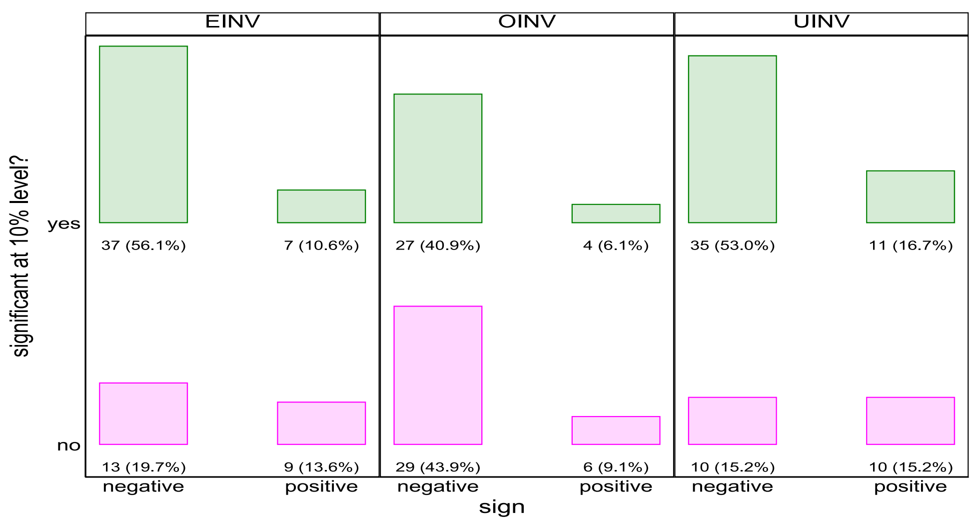

Fig. 2. Estimates of EPU on EINV, OINV, and UINV, respectively.

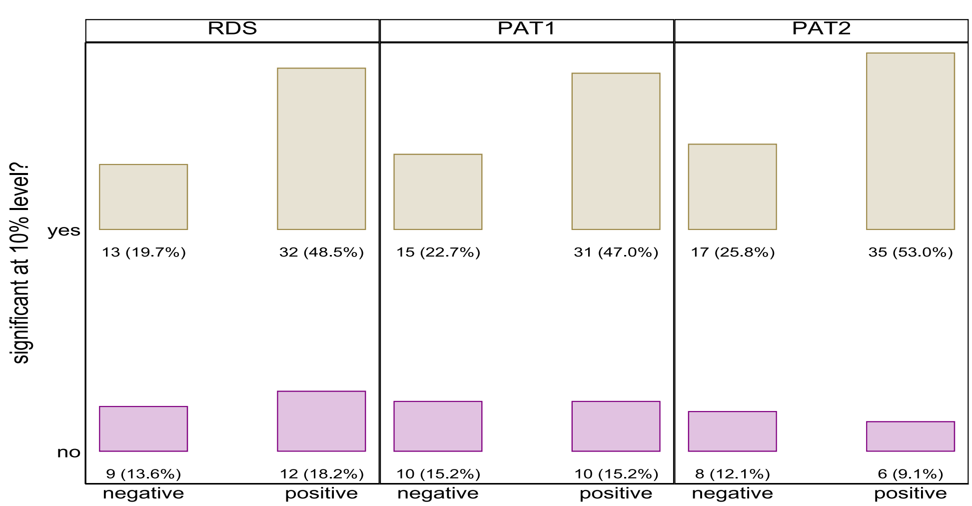

Fig. 3. Estimates of TPU1 on RDS, PAT1, and PAT2, respectively.

Long_LEV) are considered. Taking LEV as an example, in the left panel of Fig. 1, it is found that 4 estimates of OVX are statistically insignificant (2 estimates are negative, and 2 are positive). Moreover, 17 (17/55 = 30.9%) estimates are significant negative and 34 (34/55 = 61.8%) are significantly positive, all being at 10% level or less. While the probability of obtaining a significantly positive estimate is two times higher than getting a significantly negative coefficient, it cannot rule out the possibility that different authors man come out with opposite findings (negative vs positive estimates of OVX) simply because of removing different (pairs of) year dummies. Similar patterns are found for both Short_LEV and Long_LEV.

Fig. 2 illustrates the frequency/percentage of these EPU coefficients in bar graphs. Taking the overall investment inefficiency EINV as an instance (the dependent variable), it can be seen that, among 66 observations, 22 estimates are indifferent from 0 (13 negative and 9 positive estimates). In the remaining 44 statistically significant (at 10% level or less) observations, 37 (37/66 = 56.1%) are negative and 7 are positive. The distributions of signs and significances are quite similar when OINV and UINV are examined. Taken as a whole, while there is a larger chance to obtain a statistically negative estimate on EPU, there is also a (relatively smaller) probability to find a positive coefficient on EPU, depending on which year fixed effects are dropped.

Finally, Fig. 3 shows the bar graph of estimated TPU1 values across alternative dependent variables. Taking RDS as an instance, among these 66 estimates, 21 (9 are negative, and 12 are positive) are statistically insignificant. While 32 estimates are statistically significant (32/66 = 48.5%), there are still 13 coefficients (13/66 = 19.7%) being negative and statistically significant at 10% or less. Similar patterns can be found when we consider the PAT1 and PAT2 cases. Overall, while there is a higher probability to obtain statistically positive estimates, authors are also possible to get negative and statistically significant coefficients, depending on which pair of year dummies is omitted in the regression.

4. Some practical suggestions

Given the existence of perfect multicollinearity in these settings, what can we do to measure the impacts of macro time series covariates on micro panel data variables? Some potential directions are listed below.

4.1. Ignore year (time) dummies

As demonstrated above, macro variables are perfectly correlated with year dummies, but not with individual FEs. Thus, one possible way is to simply exclude year fixed effects and focus only on the firms/industries fixed effects. Moreover, when quarterly/monthly data are used for empirical analyses, we can further add quarter/month (instead of year) dummy variables (to account for seasonality). Please see Gulen and Ion (2016), Bonaime et al. (2018), Kalcheva et al. (2021), and Li and Qiu (2021) for related settings and suggestions.

4.2. Individual-variant macro-level variables

The reason that macro-level variables are perfectly correlated with year dummies is because macro-level variables are individual (firm/industry) invariant for each year. In other words, for each year, all firms/industries share the same value of the macro-level variable. Accordingly, in order to avoid the problem of perfect multicollinearity, one can try to build an individual-variant macro-level index, whenever feasible. In this respect, Yu et al. (2021) construct a novel provincial EPU index in China, and estimate the impact of EPU on manufacturing firms' carbon emission intensity.

4.3. Differential micro-level intensity/sensitivity

Since OPU, EPU and TPU are likely to influence firms/industries in a distinctive way or extent, one might want to rely upon the empirical specification of Rajan and Zingales (1998), page 563) to analyze the differential effects of OPU, EPU, and TPU on firms/industries' decisions and outcomes. In practice, it is probable that different firms/industries respond to uncertainty/shocks of oil price, economic policy and trade policy in a differential manner, i.e., some are more sensitive and others are less sensitive to a given macro-level shock. Taken as an instance, use TPU as the representative macro-level variable and consider the following specification:

$$y_{i,t} = \beta_0 + \beta_1 x_{i,t} + \beta_2 (\text{Exposure}_{i,t} \times \text{TPU}_t) + \mu_i + \mu_t + \epsilon_{i,t} \tag{4}$$

where $\text{Exposure}_{i,t}$ measures the sensitivity of firm i in year t to a given change in TPU, and TPU_t is the degree of trade policy uncertainty. Moreover, μ_i and μ_t are the firm and year fixed effects, and whenever necessary, other controlling variables can be added into the regression. The key parameter is β_2, which measures the estimated effect of an increase in TPU (say) from the 25th to the 75th percentile level on the difference in $y_{i,t}$ between the firms/industries located at the 25th and 75th percentile levels of sensitivity $\text{Exposure}_{i,t}$.

4.4. Cross-country analysis

For instance, Attig et al. (2021) use data from 19 countries and examine the impacts of EPU on dividend policy. In addition, using same countries and similar time span, El Ghoul et al. (2021) investigate how EPU affect accounting quality. Note that they both include year FEs in all tables. Since EPU for different countries are (generally) distinct in each year, it is thus not perfectly collinear with year fixed effects. Therefore, the key parameter of EPU can be identified even if the year dummies are included.[6]

5. Conclusion

There is an increasing literature using combined macro time series data with micro panel data to investigate the effects of country-level variables (oil price uncertainty, economic policy uncertainty, and trade policy uncertainty) on firm-level variables (leverage, investment inefficiencies, and innovation). Commonly, a standard two-way fixed effects estimator is implemented to obtain the empirical outcomes. However, we show that the macro-level variable is perfectly correlated with the year dummies, rendering the coefficients of interested macro-level variables unidentifiable. To deal with this situation, we offer some possible directions for analyzing the impacts of such macro (country-level) variables on micro (firm- or industry-level) panel data, and wish to see more sensible results in the near future.

CRediT authorship contribution statement

Ho-Chuan Huang: Conceptualization, Data curation, Writing – review & editing, Methodology, Software. **Xiuhua Wang:** Writing – original draft, Investigation, Supervision. **Xin Xiong:** Data curation, Software, Writing – original draft, Validation.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eneco.2022.106289.

[6] As explicitly stated in Attig et al. (2021), pages 7–8), *Our cross-country setting enables us to include year fixed effects, δ_t, in all regressions; this is not feasible in single-country studies of policy uncertainty, where all firms face the same level of country-specific policy uncertainty in a given period.*

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